

Optimization for Data Science, Autumn 2026 - Homework 1

Due at 11:59 pm on Sunday Oct/18 (Gradescope)

Your submitted solutions should be your own work. You may discuss the problems with peers and consult sources, including AI tools, but you must acknowledge the sources you use. **Write all submitted solutions from scratch, in your own words, and without assistance from an LLM during the writing process. Copying AI-generated solutions is not permitted.** You are responsible for understanding every part of your submission and may be asked to defend it in a one-on-one oral examination. If you cannot demonstrate that understanding, you may receive a zero on the assignment.

Problem 1 - Approximating functions

Consider any set $S \subseteq \mathbb{R}^n$ and twice continuously differentiable function $f: S \rightarrow \mathbb{R}$. Let $x \in S$ and $s \in \mathbb{R}^n$ be such that $x + ts \in S$ for all $t \in [0, 1]$.

(a) By defining $\theta(t) = f(x + ts)$ and using the Fundamental Theorem of Calculus:

$$\theta(1) = \theta(0) + \int_0^1 \theta'(t) dt,$$

show that

$$|f(x + s) - f(x) - \nabla f(x)^T s| \leq \frac{1}{2} L \|s\|_2^2$$

whenever f has an L -Lipschitz continuous gradient on S .

(b) Justify the formula

$$\theta(1) = \theta(0) + \theta'(0) + \int_0^1 \int_0^t \theta''(\alpha) d\alpha dt.$$

Hence, show that

$$\left| f(x + s) - f(x) - \nabla f(x)^T s - \frac{1}{2} s^T \nabla^2 f(x) s \right| \leq \frac{1}{6} Q \|s\|_2^3$$

whenever f has a Q -Lipschitz continuous Hessian on S with the operator norm (that is, when any $x, y \in S$ have $\|\nabla^2 f(x) - \nabla^2 f(y)\|_{op} \leq Q \|x - y\|_2$ where $\|M\|_{op} = \sup\{\|Mu\|_2 \mid \|u\|_2 \leq 1\}$).

Problem 2 - A calculus for the PL inequality

In Lecture 3 we proved that composition with a surjective affine map preserves PL. You may use that result without proof. This problem investigates other operations that preserve PL.

(a) **Rescaling.** Suppose that f satisfies μ -PL. For $a > 0$ and $b \in \mathbf{R}$, prove that $af + b$ is $a\mu$ -PL.

(b) **Sums in separate variables.** Let $f_i: \mathbf{R}^{d_i} \rightarrow \mathbf{R}$ be μ_i -PL, and let $a_i > 0$, for $i = 1, \dots, k$. Show that

$$F(x_1, \dots, x_k) = \sum_{i=1}^k a_i f_i(x_i)$$

is PL with constant $\min_i a_i \mu_i$. Use the Euclidean norm on the product space.

(c) **Sums in the same variables.** Construct two function in $f, g: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that each function satisfies 1-PL and they have common global minimizers, but $f + g$ does not satisfy a PL inequality with any positive constant. Explain why this does not contradict part (b).

(d) **When addition preserves PL.** Suppose that $f, g: \mathbb{R}^d \rightarrow \mathbb{R}$ are PL with constants $\mu_f, \mu_g > 0$. Assume that there exists $\gamma \in [0, 1)$ such that

$$\langle \nabla f(x), \nabla g(x) \rangle \geq -\gamma \|\nabla f(x)\| \|\nabla g(x)\| \quad \text{for every } x.$$

Prove that $f + g$ is PL with constant $(1 - \gamma) \min\{\mu_f, \mu_g\}$. A common minimizer is not assumed. Explain how the example in part (c) violates the gradient condition.

(e) **Products.** Similarly to (c), give two nonnegative, smooth functions satisfying PL whose product does not satisfy a PL inequality with any positive constant. You may take the two functions to be identical. Prove both the PL property of the factors and the failure for their product.

(f) **PL does not imply convexity, even in one dimension.** Construct a twice continuously differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfies a global PL inequality but is not convex. Give an explicit positive PL constant and justify nonconvexity. A globally Lipschitz gradient is not required.

Problem 3 - Why regularize an interpolating model?

Fix $X \in \mathbb{R}^{n \times d}$ with $d > n$ and $\text{rank}(X) = n$. Suppose that the observations are

$$y = s + \varepsilon, \quad s \in \mathbb{R}^n, \quad \mathbf{E}[\varepsilon] = 0, \quad \mathbf{E}[\varepsilon \varepsilon^\top] = \sigma^2 I_n, \quad \text{and} \quad \sigma > 0.$$

The design X and the noiseless response vector s are fixed. Expectations are over the observation noise. For an estimator $\hat{s}$ of s , define its “prediction risk” at the observed inputs by

$$\mathcal{R}(\hat{s}) = \frac{1}{n} \mathbf{E} \|\hat{s} - s\|^2.$$

The questions below concern the statistical consequences of ridge regression.

(a) **Bias and variance.** For a deterministic matrix $B \in \mathbb{R}^{n \times n}$, consider the linear estimator $\hat{s} = By$. Prove that

$$\mathcal{R}(By) = \underbrace{\frac{1}{n} \|(B - I_n)s\|^2}_{\text{squared bias}} + \underbrace{\frac{\sigma^2}{n} \text{tr}(BB^\top)}_{\text{variance}}.$$

Suppose we can write $B = U \text{diag}(b_1, \dots, b_n) U^\top$, where U is orthogonal with columns u_j . Express both terms as sums involving b_j and $u_j^\top s$.

(b) **The ridge prediction filter.** For $\alpha > 0$ fixed independently of the noise, let w_α be the ridge estimator from Lecture 3, with penalty $\frac{\alpha}{2} \|w\|^2$. Using the formula derived in class, write $Xw_\alpha = B_\alpha y$. Let $K = \frac{1}{n} X X^\top$, and choose an orthogonal matrix U whose columns satisfy $K u_j = \lambda_j u_j$. Determine the eigenvalues of B_α in this basis. Which components of the observed responses are shrunk most strongly toward zero?

(c) **The statistical tradeoff.** Use part (a) to derive an explicit formula for

$$R(\alpha) := \frac{1}{n} \mathbf{E} \|Xw_\alpha - s\|^2$$

as a sum of squared bias and variance in the eigenbasis of K . Show that increasing α increases the squared bias and decreases the variance, where “increases” and “decreases” allow equality. Compute the limits as $\alpha \downarrow 0$ and $\alpha \rightarrow \infty$. Also compute the risk of any estimator $\hat{w}$ satisfying $X\hat{w} = y$ exactly.

(d) **Less accurate fitting can give more accurate estimation.** Define $R(0)$ by the limit in part (c). Prove that, for every fixed s , there exists $\alpha > 0$ such that $R(\alpha) < R(0)$. Explain why this result motivates ridge regression even when the training data can be fit exactly. Does it imply that every positive value of α improves prediction risk? Justify your answer.

Problem 4 - Early stopping as spectral filtering

Consider the same setup as in Problem 3, now gradient descent on the unregularized objective

$$f(w) = \frac{1}{2n} \|Xw - y\|^2, \quad \text{i.e.,} \quad w_{k+1} = w_k - \frac{\eta}{n} X^\top (Xw_k - y), \quad \text{and} \quad w_0 = 0,$$

where $0 < \eta \leq 1/L$ and $L = \lambda_{\max}(\frac{1}{n}XX^\top)$. A common strategy in data science is to stop gradient descent early since it often leads to useful statistical properties of its output. The following questions develop the spectral interpretation of early stopping.

(a) **Each direction has its own learning speed.** Using the recursion for the residual $Xw_k - y$ proved in Lecture 3, find the scalar function $g_k(\lambda)$ such that

$$u_j^\top Xw_k = g_k(\lambda_j) u_j^\top y.$$

The multiplier $g_k(\lambda)$ is called a *spectral filter*. Show that, on $0 \leq \eta\lambda \leq 1$, it takes values in $[0, 1]$ and is nondecreasing in both k and λ .

(b) **Which directions have been learned?** For every integer $k \geq 1$, prove

$$1 - e^{-\eta k \lambda} \leq g_k(\lambda) \leq \min\{1, \eta k \lambda\}.$$

Use these bounds to explain the regimes $\eta k \lambda \ll 1$ and $\eta k \lambda \gg 1$. Compare this filter with the ridge filter from Problem 3(b). Explain why $\alpha \approx 1/(\eta k)$ matches their transition scales. Are GD at finite time and ridge with $\alpha = 1/(\eta k)$ the same estimator? Justify your answer.

(c) **Avoiding unstable inversion.** Write the thin SVD as $X = U \text{diag}(\rho_1, \dots, \rho_n) V^\top$, where V has orthonormal columns v_j and $\rho_j^2 = n\lambda_j$. Use part (a) to show that

$$w_k = \sum_{j=1}^n \frac{g_k(\rho_j^2/n)}{\rho_j} (u_j^\top y) v_j.$$

You may use the fact proved in Lecture 3 that zero-initialized GD stays in the range of $X^\top$. Then use part (b) to prove that the response-to-coefficient multiplier satisfies

$$0 \leq \frac{g_k(\rho_j^2/n)}{\rho_j} \leq \min \left\{ \frac{1}{\rho_j}, \frac{\eta k \rho_j}{n} \right\}.$$

Contrast its behavior as $\rho_j \downarrow 0$, with η and finite k fixed, with the multiplier $1/\rho_j$ in the minimum-norm interpolator.

- (d) **Optimization error versus prediction risk.** For a fixed iteration count k , chosen independently of the observation noise, use part (a) and Problem 3(a) to derive formulas, as sums over the eigendirections, for both

$$f(w_k) = \frac{1}{2n} \|Xw_k - y\|^2 \quad \text{and} \quad R_k = \frac{1}{n} \mathbf{E} \|Xw_k - s\|^2.$$

Show that training loss is nonincreasing along every run. Show that the squared bias in R_k is nonincreasing while its variance is nondecreasing, and compute R_0 and $\lim_{k \rightarrow \infty} R_k$. Explain why decreasing training loss need not imply decreasing prediction risk. How does early stopping differ from the minimum-norm implicit bias established in Lecture 3?

For parts (e)–(f), use Python and NumPy with the following setup. Here K is diagonal, so $U = I_n$, and the signal lies in its first four eigendirections.

```
import numpy as np

rng = np.random.default_rng(31019)
n, d = 20, 40
eigenvalues = np.geomspace(1.0, 1e-3, n)
V, _ = np.linalg.qr(rng.standard_normal((d, n)), mode="reduced")
X = np.sqrt(n) * (np.sqrt(eigenvalues)[:, None] * V.T)
signal = np.zeros(n)
signal[:4] = 3.0
sigma, eta = 1.0, 1.0
steps = np.array([0, 1, 2, 3, 5, 10, 20, 50,
                  100, 300, 1000, 3000])
```

- (e) **Implement and check GD.** Generate one observation vector $y = s + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$, and implement GD from scratch for 3000 iterations. Keep this y fixed throughout the run. You may use NumPy for matrix operations, but not a library optimization routine. At the iteration counts listed above, compare Xw_k with the prediction obtained from your spectral formula in part (a), and report the largest discrepancy in Euclidean norm. Plot the training loss and the realized signal prediction error $\|Xw_k - s\|^2/n$ against k .
- (f) **Compare the experiment with the theory.** Generate 500 independent noise vectors with the same distribution, keeping X and s fixed. For each listed iteration count, estimate R_k by averaging the signal prediction error over these observations. You may use the spectral formula rather than running GD separately for every observation. Plot the empirical risks together with the exact theoretical risks from part (d), their squared-bias and variance terms, and the risk of exact interpolation. Use an axis scale that makes the early iterations visible. Identify a finite iteration count whose expected risk is smaller than that of interpolation. Explain what the experiment illustrates about fitting signal and noise, and why knowledge of s in this experiment does not by itself supply a stopping rule for real data, where s is unknown.